Kaustubh Shandilya

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Education

BMS Institute of Technology and Management — Bengaluru, Karnataka
Bachelor of Engineering in Computer Science (AI/ML) — *Oct 2024 – Present*

- Current CGPA: 9.15 / 10.0
- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Management Systems, Statistics & Probability, Linear Algebra

Projects

Sentinel – RAG-Based Attack Surface Management Tool | *FastAPI, ChromaDB, LlamaIndex, Ollama, Scikit-Learn* 2025

- Architected a production-ready RAG pipeline over live network assets using ChromaDB and LlamaIndex, enabling sub-second natural language querying through a REST API – replacing manual asset lookup.
- Built a multi-page Streamlit dashboard surfacing real-time risk distribution and vulnerability heatmaps, cutting triage time for security decisions.
- Deployed a Random Forest risk-scoring model across 13 engineered features, enabling automated asset prioritization without human intervention.

DataLens – AI-Powered EDA Automation Tool | *FastAPI, Streamlit, Groq LLM (Llama 3.3 70B), Supabase* 2025

- Reduced dataset profiling from hours to seconds by automating target detection, missing value analysis, outlier flagging, and class imbalance checks via an LLM-orchestrated pipeline.
- Surfaced feature importance rankings using Random Forest, giving users an instant signal on which variables drive predictive value – without writing a single line of analysis code.
- Auto-generated a full visual report suite (correlation heatmaps, box plots, histograms, pie charts) tailored to each dataset's schema and data types.

Movie Recommender System | *Python, FastAPI, Streamlit, Scikit-Learn, MovieLens 100K* 2025

- Benchmarked content-based, collaborative filtering, and hybrid models on MovieLens 100K; hybrid approach achieved a **20x accuracy gain** over baseline (Precision@10: $0.007 \rightarrow 0.145$).
- Ran a controlled A/B test between two algorithms and validated the performance gap was statistically significant ($p < 0.001$), applying hypothesis testing to a real-world ML evaluation.
- Measured popularity bias via Gini coefficient and catalog coverage, documenting the precision-diversity tradeoff – an insight directly applicable to production recommender design.

Technical Skills

Languages: Python, C/C++, SQL, PostgreSQL, JavaScript, HTML/CSS
Frameworks & Tools: FastAPI, Streamlit, Next.js, LlamaIndex, ChromaDB
Machine Learning: Scikit-Learn, Pandas, NumPy, Matplotlib, Random Forest, Collaborative Filtering, RAG
Developer Tools: Git, GitHub, VS Code, Supabase, Vercel, Render, GitHub Copilot
Areas of Interest: Data Science, Machine Learning, Exploratory Data Analysis, Recommender Systems, NLP

Leadership & Community

Event Coordinator — BMS Institute of Technology and Management
AWS Builder Community – College Chapter — *Mar 2026 – Present*

- Plan and execute cloud and AI/ML-focused technical events and workshops, driving hands-on learning for 100+ students across the college chapter.
- Bridge the gap between AWS ecosystem tools and student projects by facilitating peer knowledge-sharing sessions on cloud infrastructure and deployment.